PIPELINE AUDIT AND HEALTH REPORT

Deadlock demo: a rework loop with no slack

Press and Check feed each other through bounded buffers. When both fill, each waits for the other forever. The simulator detects it and prints the wait-for cycle. Fix: add "bypass": true to the rework edge (or set a capacity of -1 on one node) and run again.

Generated 2026-10-13 19:48 | simulated 2.0 h | seed 3 | 10 replications

HEALTH SCORE

24 /100

Grade F

Not fit for use: the pipeline deadlocks.

Critical 1 | High 2 | Medium 2 | Low 2 | Info 1

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
238.4 units/h	37 units	48.2%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
3.2 units	43 s	press

Executive summary

DEADLOCK at t=558.7s: every remaining node is waiting on another. Wait-for cycle: press -> check -> press. Break the loop with a rework bypass lane ("bypass": true) or an unbounded buffer.

What to fix first

- Critical (press -> check -> press): Break the circular wait: put "bypass": true on the rework edge, give one node in the loop an unbounded buffer, or route rework to its own lane.
- High (arrivals): Add capacity at the constraint or enlarge the buffer in front of it. Measured best lever: +2 buffer slots at 'check' (+39.2% throughput).
- High (press): Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: +2 buffer slots at 'check' (+39.2% throughput).
- Medium (steady state): Lengthen the run or use a warm-up period before trusting averages.

Best measured improvement

Best lever: +2 buffer slots at 'check' -> +70.1 units/hour (39.2%), measured over 3 seeds with common random numbers.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
Critical	Integrity	**press -> check -> press**	The pipeline deadlocks at t=559 s: every remaining node waits on another (wait-for cycle: press -> check -> press). Action: Break the circular wait: put "bypass": true on the rework edge, give one node in the loop an unbounded buffer, or route rework to its own lane.	-40
High	Flow	**arrivals**	48.2% of arrivals are lost because a buffer is full. Action: Add capacity at the constraint or enlarge the buffer in front of it. Measured best lever: +2 buffer slots at 'check' (+39.2% throughput).	-15
High	Capacity	**press**	The bottleneck is 98% utilised (busy or down). There is no slack: any disturbance directly cuts output. Action: Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: +2 buffer slots at 'check' (+39.2% throughput).	-8
Medium	Model	**steady state**	Little's Law differs by 12.7%: the run has not reached steady state (units are still accumulating). Action: Lengthen the run or use a warm-up period before trusting averages.	-4
Medium	Confidence	**throughput**	Throughput varies strongly between runs: 95% CI is +-12.6% of the mean over 10 runs. Action: Run longer, use a warm-up period, or reduce variability at the bottleneck.	-4
Low	Reliability	**failure model**	No machine has a failure model (MTBF/MTTR), so the results are optimistic. Action: Add realistic MTBF and MTTR to the stations that break down in practice.	-3
Low	Model	**run length**	The measured window (0.2 h) is only 13x the cycle time. Action: Aim for at least 30x the cycle time so averages settle.	-2
Info	Reliability	**dependencies**	2 stations lie on every path (serial structure), but none has a failure model. Action: Fine for an ideal analysis; add failure data to find out how fragile this is.	

Performance by node

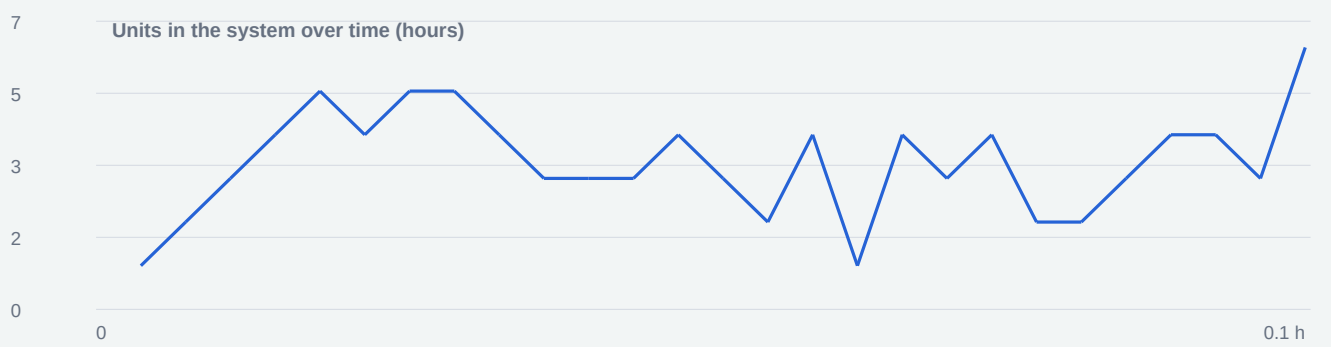
Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
source	source	-	-	-	-	-
press	machine		97.7	0.0	0.0	2.4 / 3
check	machine		54.4	6.0	0.0	0.9 / 3

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
ship	sink	-	-	-	-	-

Work in process over time



Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	208.81 units/h	+- 26.36	36.85	12.6%
Average WIP	3.32 units	+- 0.14	0.19	4.2%
Cycle time	44.03 s	+- 3.83	5.36	8.7%
Arrivals lost	47.73 %	+- 6.20	8.66	13.0%

10 of 10 replications deadlocked.

Failure model

No machine has a failure model.

Single points of failure

All units must pass through: press, check. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 83 = completed 37 + scrapped 0 + lost 40 + in system 6
Little's Law	FAIL	L = 3.23, lambda x W = 2.82 (gap 12.7%)
Run length	WARN	window = 13 x cycle time (30x or more recommended)
Warm-up	WARN	no warm-up: start-up transients are included in averages
Deadlock-free	FAIL	deadlock at t = 559 s

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 178.7 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
check	+2 buffer slots	248.9	**+70.1**	+39.2%
check	10% faster processing	199.2	**+20.4**	+11.4%
press	+2 buffer slots	178.7	**+0.0**	+0.0%
press	10% faster processing	141.2	**-37.5**	-21.0%

Best lever: +2 buffer slots at 'check' -> +70.1 units/hour (39.2%), measured over 3 seeds with common random numbers.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
source	source	interarrival=exponential(7)
press	machine	capacity=2, process=normal(10,1)
check	machine	capacity=2, process=exponential(6)
ship	sink	(defaults)

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
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High	**arrivals**	48.2% of arrivals are lost because a buffer is full.	-15
High	**press**	The bottleneck is 98% utilised (busy or down). There is no slack: any disturbance directly cuts output.	-8
Medium	**steady state**	Little's Law differs by 12.7%: the run has not reached steady state (units are still accumulating).	-4
Medium	**throughput**	Throughput varies strongly between runs: 95% CI is +-12.6% of the mean over 10 runs.	-4
Low	**failure model**	No machine has a failure model (MTBF/MTTR), so the results are optimistic.	-3
Low	**run length**	The measured window (0.2 h) is only 13x the cycle time.	-2

SEVERITY	WHERE	FINDING	POINTS
	Total deductions		**-76**
	Health score		**24**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.